



Accurate Name Extraction from News Video Graphics

Andrea Filiberto Lucas | B.Sc. ICT (AI) Dissertation
Supervised by Dr. Dylan Seychell



L-Università ta' Malta
Faculty of Information &
Communication Technology

Department
of Artificial
Intelligence

Introduction

Quick Introduction...

Problem Definition:

Extract personal names embedded within graphical elements (such as lower-thirds, breaking news banners, and headlines) in news video content.

Challenges:

Names are visually embedded in graphics rather than spoken or subtitled, making them harder to detect [4–7].

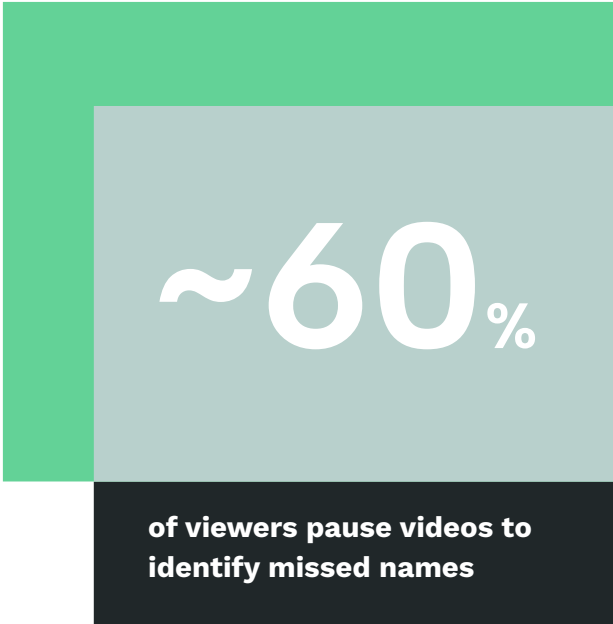
Graphical styles vary across broadcasters, lacking consistent placement, typography, or colour schemes [13, 14], thus complicates both visual detection and text extraction.

Manual extraction is slow and error-prone.

Why This Matters?

Names embedded in news graphics **provide essential context** for understanding who is involved in a story [18, 19].

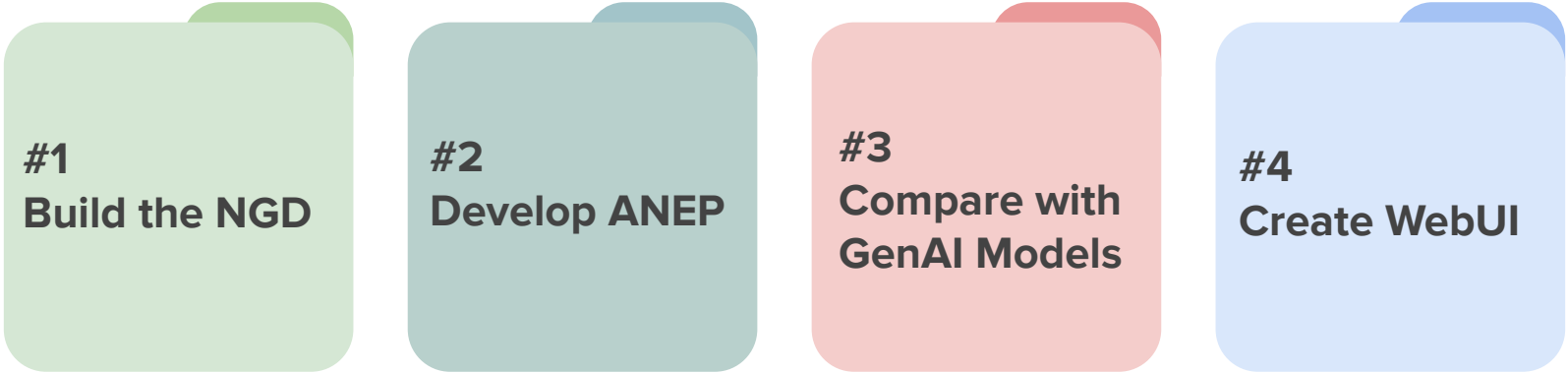
Current media analysis tools **ignore visual overlays entirely** [8–12], leaving a major gap in automated understanding of video content.



~60%

of viewers pause videos to
identify missed names

Objectives of the Dissertation



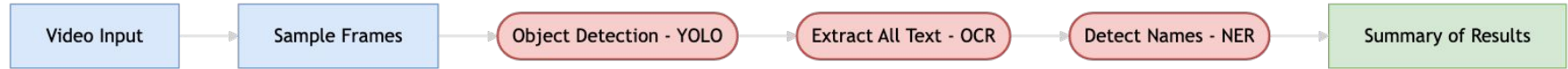
#1
Build the NGD

#2
Develop ANEP

#3
**Compare with
GenAI Models**

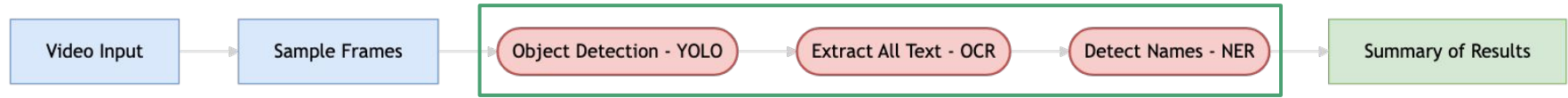
#4
Create WebUI

The Accurate Name Extraction Pipeline (ANEP)



#2
Develop
ANEP

The Accurate Name Extraction Pipeline (ANEP)



#2
Develop
ANEP

3 Main Components



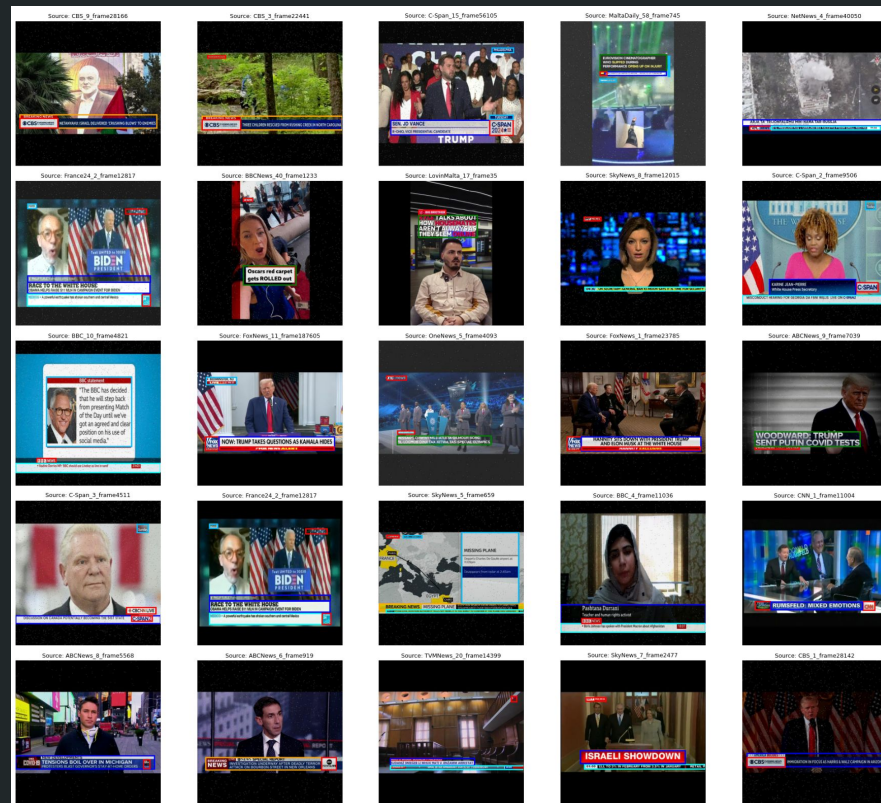
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Methodology

The NGD

(News Graphics Dataset)

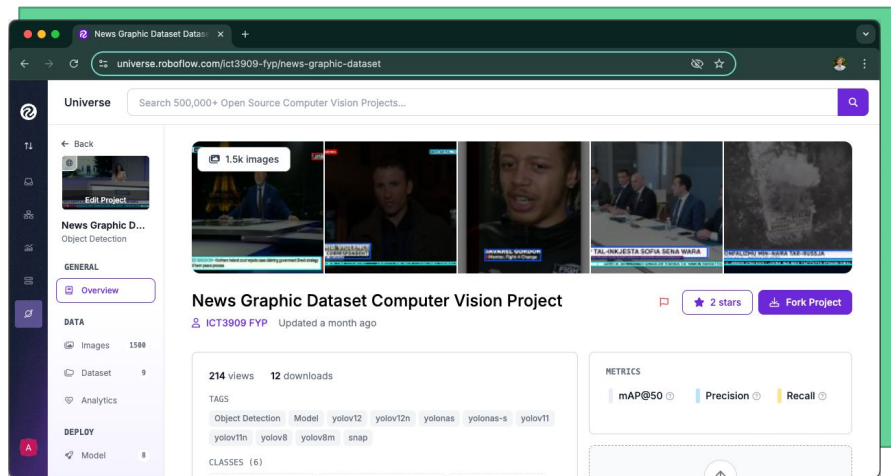


Building the NGD

1,500 carefully annotated frames from diverse news sources.

Sources span traditional TV and social media-based news platforms, from both local and international broadcasters.

Six distinct graphic types identified and labeled.

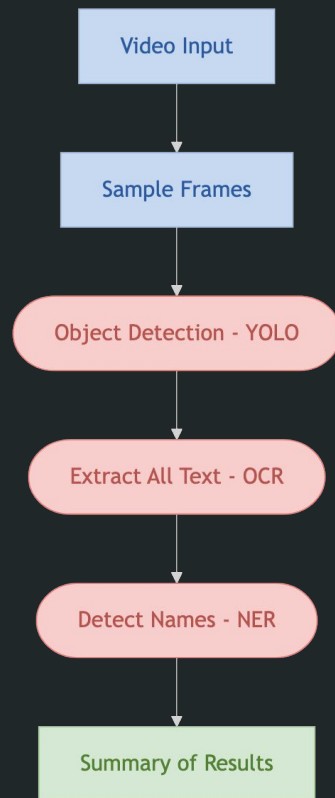


(Dataset was uploaded and published onto Roboflow)



The ANEP

(Accurate Name Extraction Pipeline)



ANEP Architecture

1. Frame Processing

Sample video frames and remove redundancy using multi-level deduplication.

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Detect graphical overlays using YOLOv12 OD and preprocess cropped ROIs.

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Extract text from preprocessed ROIs using Tesseract OCR.

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4. Named Entity Extraction & Validation

Apply ensemble NER models (spaCy with GLiNER & BERT) to validate names.

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5. Name Clustering & Timeline Generation

Cluster similar names, select canonical forms, and generate appearance timelines.

What is Object Detection (OD)?



OD is the process of **automatically locating and classifying ROIs** [22].

Avoids processing the whole frame unnecessarily.

Improves text **accuracy by reducing noise** in extracted names.

What is Optical Character Recognition (OCR)?

A spokesperson for the ITV show said:
"Mickey Rourke has agreed to leave
the Celebrity Big Brother House this
evening following a discussion with
Big Brother regarding further use of

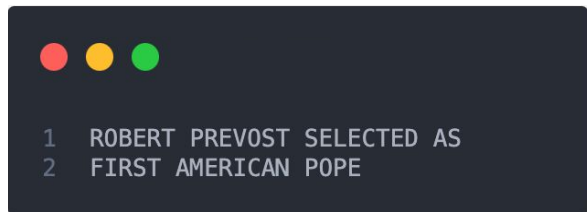
```
1 A spokesperson for the ITV show said:
2 "Mickey Rourke has agreed to 'eave |
3 4 the Celebrity Big Brother this ,
4 | evening following a discussion with
5 i Big Brother regarding further use of
```

OCR is the process of **converting images of text into machine-readable text** [31, 32].

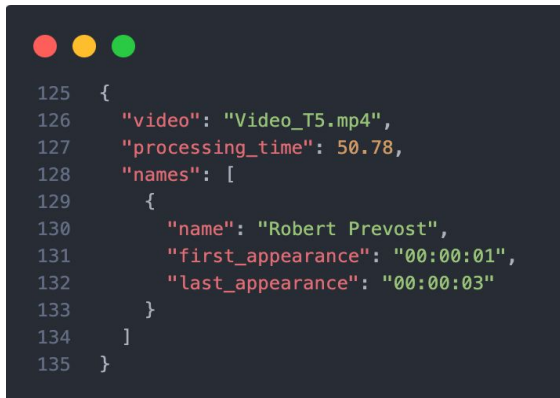
ANEP uses **Tesseract OCR** to extract text from graphical regions.

Enables accurate name recognition by **feeding clean text into the NER** stage.

What is Name Entity Recognition (NER)?



```
1 ROBERT PREVOST SELECTED AS  
2 FIRST AMERICAN POPE
```



```
125 {  
126   "video": "Video_T5.mp4",  
127   "processing_time": 50.78,  
128   "names": [  
129     {  
130       "name": "Robert Prevost",  
131       "first_appearance": "00:00:01",  
132       "last_appearance": "00:00:03"  
133     }  
134   ]  
135 }
```

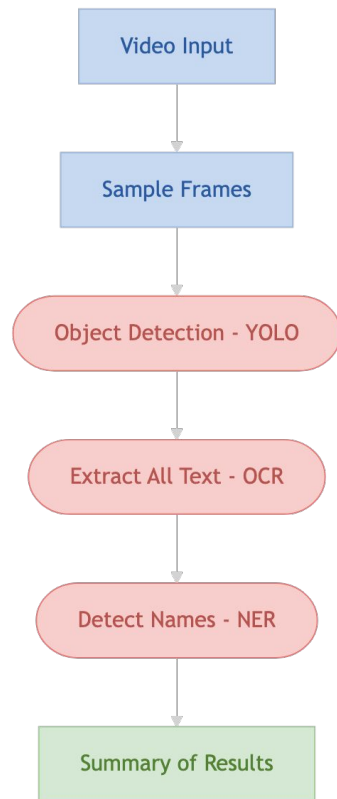
NER is the process of **identifying and classifying names** and entities **in text** [35, 36].

ANEP uses spaCy with GLiNER, & Transformer-based BERT.

Zero-shot models can **recognise entities without task-specific training** [75].

Advantages of the Architecture

1. **Modular design** enables **component-level tuning** and replacement.
2. **Explainable outputs** provide clear, traceable audit trails.
3. **Adaptable framework** **supports diverse graphic styles** and news formats.
4. **Transparent processing** allows for **effective debugging** and iterative **refinement**.



GenAI

(Generative Artificial Intelligence)



GenAI for Comparison & Benchmarking

1. Google Vision API + Gemini 1.5 Pro

Combines robust OCR with powerful multimodal reasoning over visual content [83–85].

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2. LLaMA 4 Maverick

Uses sparse MoEs and iRoPE attention for efficient multimodal understanding [86–88].

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3. Why it matters?

Both offer cutting-edge, general-purpose extraction [89].

BUT requires high external compute and careful prompting.

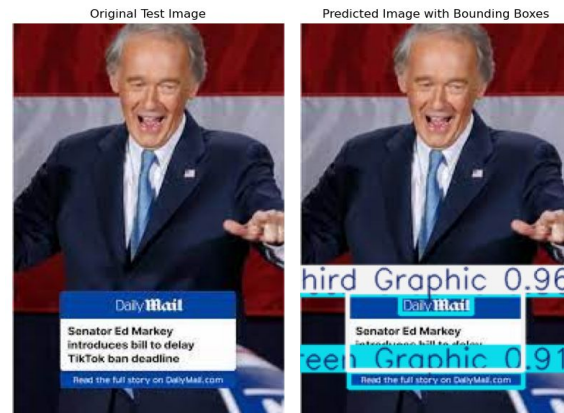
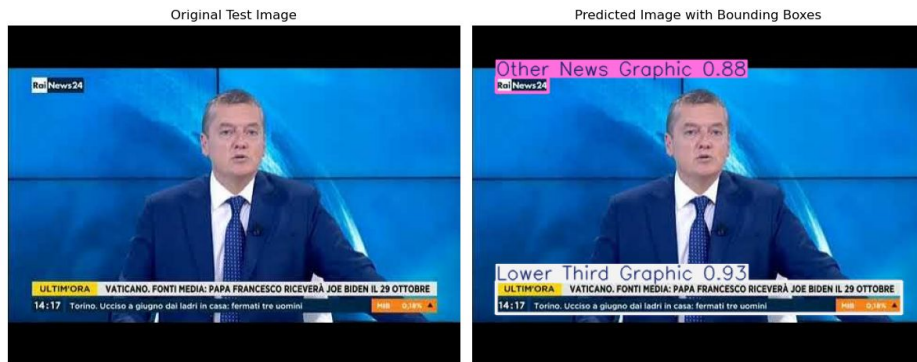


Evaluation of Results

Comparison of Yolo-based OD models trained on NGD

Model	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Epochs
Locally Trained Models					
Yolov12(m)	93.9%	93.5%	95.8%	88.7%	102
Yolov8(m)	92.6%	86.9%	93.7%	75.2%	47
Roboflow Platform Models					
Yolov12(n)	91.6%	90.8%	93.8%	85.4%	120
Yolov11(n)	91.2%	90.4%	93.1%	84.9%	100
Yolov12(n) (Reflect)	91.4%	85.7%	91.8%	80.4%	72
Yolo-NAS(n)	85.1%	84.3%	91.0%	61.0%	51

News Graphics OD Key Insights



Best performance from locally trained **YOLOv12(m)**.

Achieved high accuracy on well-structured graphics such as lower-thirds and tickers, while performance declined on the “Other” category due to visual ambiguity.

Model **generalised well across unseen** broadcasters & social media formats.

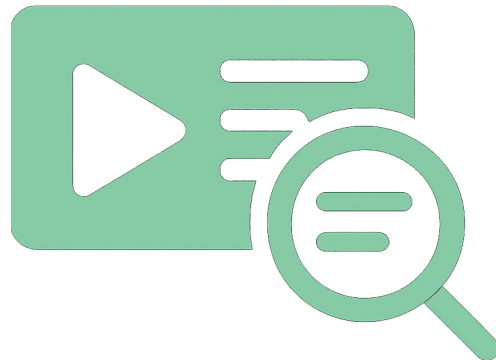
ANEP Evaluation Setup

Same input videos used across all pipelines to ensure fair comparison.

Standard metrics applied: Precision, Recall, and F1-score.

Processing time measured for each system to assess efficiency.

Enables direct comparison between traditional and GenAI-based (multimodal) approach.



Performance of Name Extraction Models

Model	Precision (%)	Recall (%)	F1 Score (%)	Time (seconds)
ANEP	79.92	74.44	68.10	542.15
GVA & Gemini 1.5 Pro	93.33	76.67	82.22	94.68
LLaMA 4 Maverick	66.67	50.00	140.18	140.18

-  **Gemini 1.5 Pro:** Highest F1, fast, strong accuracy – best for real-time use.
-  **ANEP:** Slower but explainable – ideal for traceable contexts.
-  **LLaMA 4:** Simple and cost effective – struggles with OCR noise and ambiguity.

Limitations of the Dissertation

1. Dataset & Visual Challenges:

NGD coverage excludes some regional styles and non-Latin scripts.

2. ANEP:

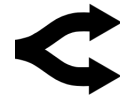
NER errors with ambiguous names, surnames without context, and accented Maltese names.

Alias resolution struggles with spelling variants, OCR noise, and informal titles.

3. LLaMA 4:

Context window limits long video processing; needs pre-tokenized input.

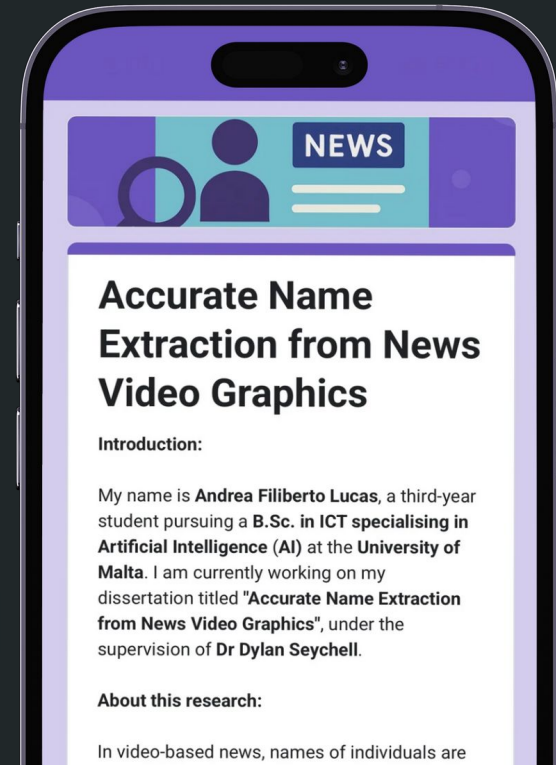
Rate limits and latency, under basic-tier use.



OpenRouter

The Survey

(aflucas.com/dissertation-form)



Accurate Name Extraction from News Video Graphics

Introduction:

My name is **Andrea Filiberto Lucas**, a third-year student pursuing a **B.Sc. in ICT specialising in Artificial Intelligence (AI)** at the **University of Malta**. I am currently working on my dissertation titled "**Accurate Name Extraction from News Video Graphics**", under the supervision of **Dr Dylan Seychell**.

About this research:

In video-based news, names of individuals are

Survey Overview

Why Conduct a Survey?

To understand **real-world news consumption habits** and **validate the design decisions behind ANEP**.

Focused on **how viewers interact with name graphics**, and **perceive the usefulness** of automated extraction tools.

Survey Details:

Total of **404** participants.

Conducted from **10 April to 10 May 2025**.

Majority aged **18–24** or **45–54**.

92.1% of the participants **reside in Malta**.

Key Findings

~82%

rely on video graphics to
identify people

~58%

paused or rewound to catch
a name they missed

~63%

trust depends on transparency
& developer reputation



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End-to-End Pipeline Demo

Step: 1 - Sample Frame



Step: 2 - Object Detection



Step: 3 - Cropping ROIs



1



2

News Headline - Conf: 0.96 Digital On-Screen Graphic - Conf: 0.89

ROBERT PREVOST SELECTED AS
FIRST AMERICAN POPE



Step: 4 - Final JSON Output



News Headline - Conf: 0.96 Digital On-Screen Graphic - Conf: 0.89

**ROBERT PREVOST SELECTED AS
FIRST AMERICAN POPE**

FOX NEWS



```
125 {
126   "video": "Video_T5.mp4",
127   "processing_time": 50.78,
128   "names": [
129     {
130       "name": "Robert Prevost",
131       "first_appearance": "00:00:01",
132       "last_appearance": "00:00:03"
133     }
134   ]
135 }
```

Interactive Demo: ANEP-UI



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Future Work & Conclusion

Future Development

Expanding the NGD to include multilingual content, including right-to-left scripts and non-Latin alphabets, would improve global coverage.

Developing multimodal systems that combine visual, text, and audio inputs could help match spoken names with on-screen graphics [90-92].

Optimising ANEP for real-time use through pruning, quantisation, and edge deployment would make it faster and more efficient for live broadcasts [93,94].



Summary of Contributions

1. The NGD:

A new, **annotated dataset capturing six graphic types** across **TV & social media** formats, now publicly available.

2. The ANEP:

A robust, **modular pipeline** integrating **YOLOv12**, domain-specific **OCR**, & **dual-layer NER** (spaCy & GLiNER/Transformer).

3. Comparative Evaluation:

Benchmarked ANEP against Gemini and LLaMA 4 pipelines using precision, recall, F1, and runtime.

4. The ANEP-UI:

A platform for uploading videos, selecting models, and visualising results.

5. The Survey:

A 404-person survey validating the **practical relevance** of the dissertation & **guiding system design**.

Key Findings & Implications

YOLOv12 detection model generalised well across noisy, unseen footage (*achieving 95.8% mAP@0.5*).

Gemini pipeline achieved highest name extraction scores (*F1: 82.22%*) and low latency.

ANEP provided explainable, balanced results (*F1: 68.10%*), more robust in traceability-focused contexts.

LLaMA 4 underperformed due to context limitations and OCR noise sensitivity.

The **survey confirmed widespread reliance** on name graphics, common **issues with readability**, and strong **support for transparent automated extraction tools**.

These results confirm that accurate, explainable, and efficient name extraction from video graphics is both: feasible & socially relevant.



Thank You

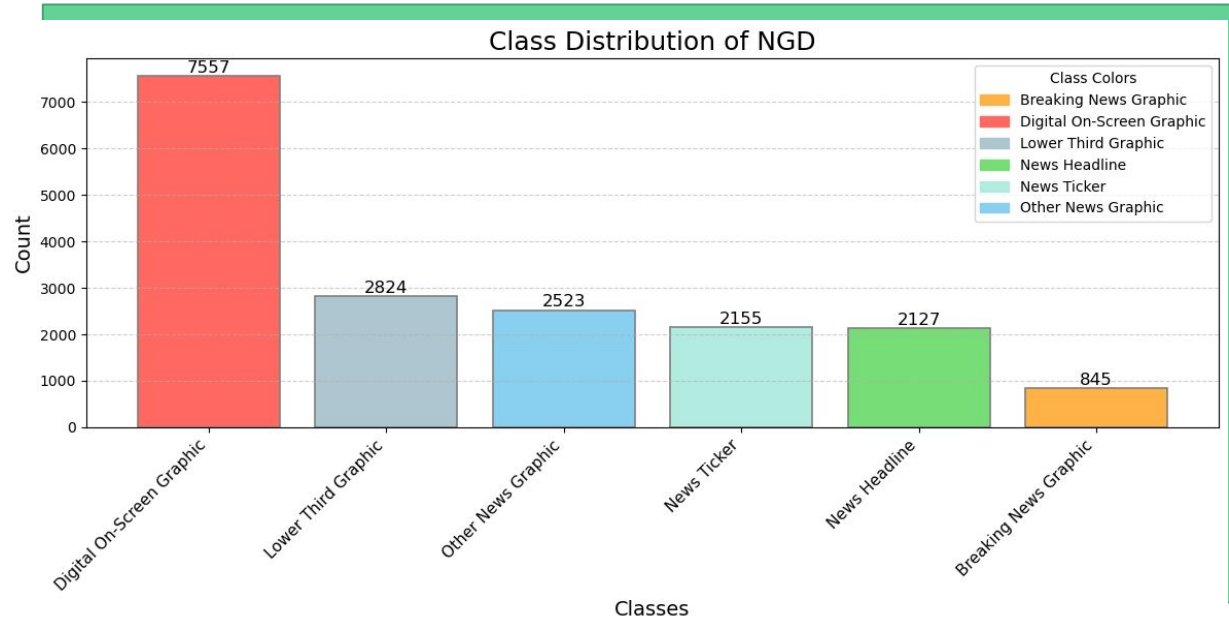
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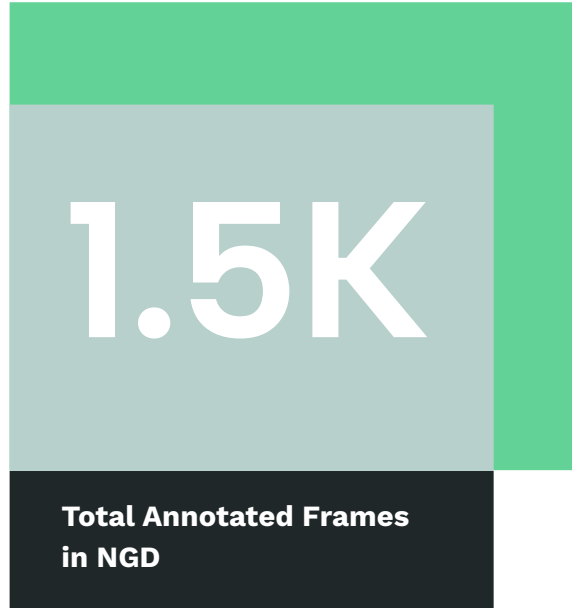
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Class Distribution of NGD



Annotated Frames by Source

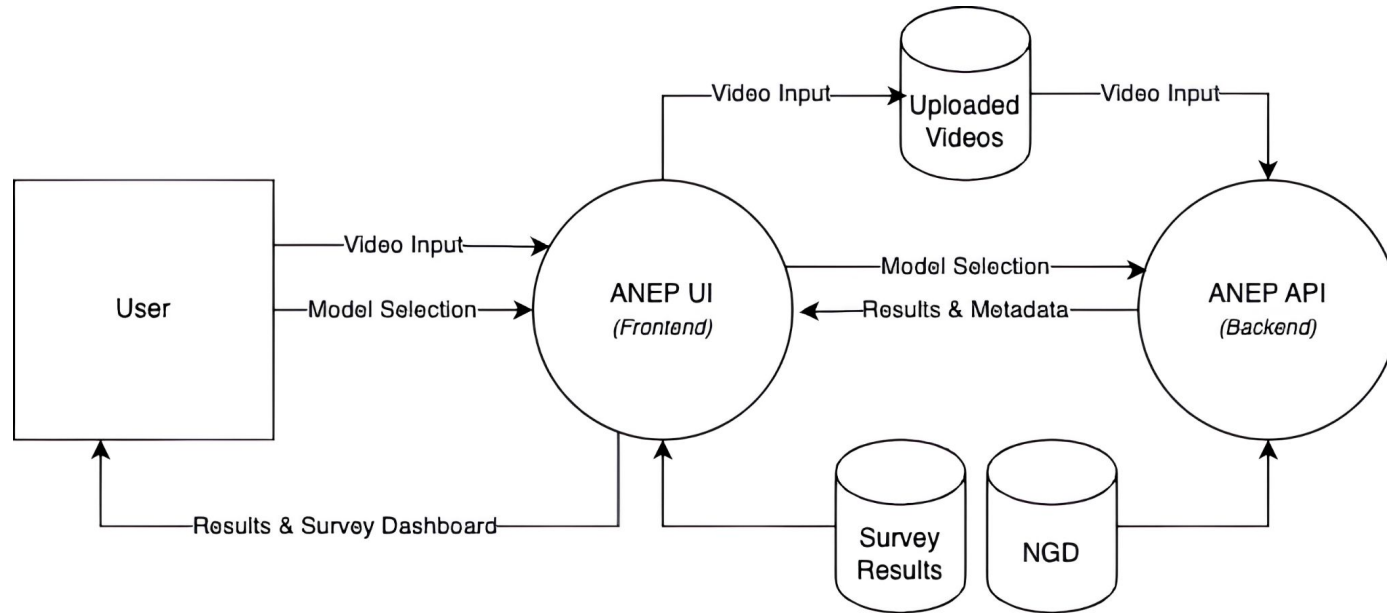


TikTok Sources	Annotated Frames
SideStreet	75
MaltaDaily	72
LovinMalta	74
DailyMail	74
SkyNews (TikTok)	74
TVMNews (TikTok)	75
BBCNews (TikTok)	75
FoxNews (TikTok)	75
Total (TikTok)	594

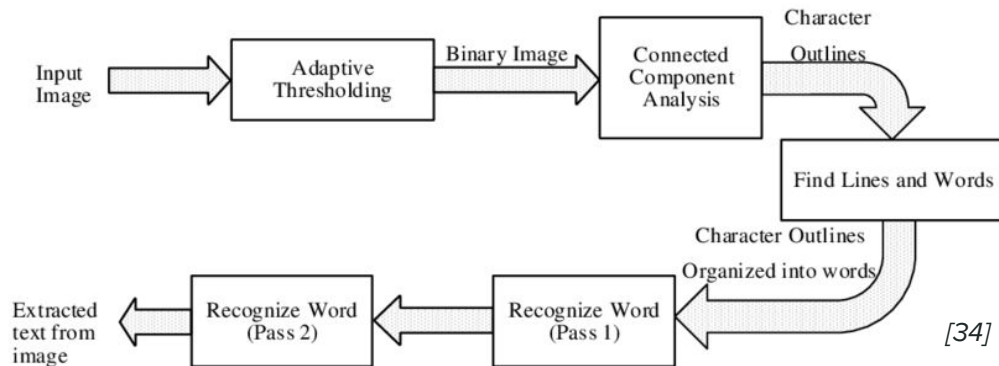
Foreign News Sources	Annotated Frames
C-SPAN	75
CNN	75
BBC	78
France24	75
ABCNews	78
CBS	75
FoxNews (Broadcast)	75
SkyNews (Broadcast)	75
Total (Foreign)	606

Local News Sources	Annotated Frames
TVM News	102
One News	97
Net News	101
Total (Local)	300

Level 0 DFD of ANEP-UI



Tesseract OCR for News Video Graphics

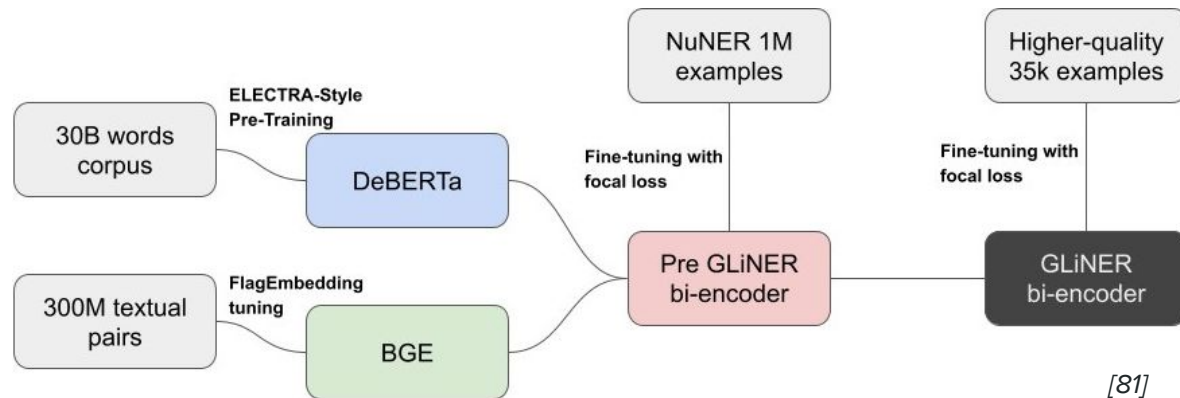


Open-source OCR engine, enhanced by Google, used to extract text from video frames using line segmentation and adaptive recognition [60].

Helps extract overlaid text despite challenges like blur, animation, and low resolution [59], [61], [63].

Struggles with video-specific noise. Improved via preprocessing (CLAHE, thresholding), dual-pass recognition, and hybrid pipelines with NER and DL models [31], [32], [66].

Overview spaCy & GLiNER



Fast NLP library with efficient NER; great for lightweight pipelines and domain-specific tuning [36].

GLiNER handles multilingual, zero-shot NER; BERT adds strong context-aware recognition for complex or noisy input [37], [75].

Both help extract real names from noisy OCR text, even in news broadcasts with animations or visual clutter [74], [76].

Prompt Engineering for Name Extraction

Gemini Prompt:

Extracts real people's names from OCR text, ignoring brands, apps, and non-name text.

LLaMA Prompts:

Uses a primary and fallback prompt to extract all visible text from video frames.

JSON Format:

Both systems return structured name data (*with timestamps*) using a strict JSON format.

```
222 prompt = f"""
223 I need to extract proper names (people's names) from the text data from a video.
224
225 The text data is from multiple frames and is as follows:
226
227 {json.dumps(text_data, indent=2)}
228
229 Analyze the text and extract ONLY real people's names. Skip:
230 - Brand names
231 - Channel names
232 - Show names
233 - App names
234 - Non-name text
235
236 Return the result as a JSON object with this exact format:
237 {{
238   "names": [
239     {{
240       "name": "Full Name",
241       "first_appearance": "timestamp",
242       "last_appearance": "timestamp",
243       "count": number_of_appearances
244     }}
245   ]
246 }}
247
248 Only include valid people's names. Return an empty array if no real people's names are found.
249 Be precise and return ONLY the JSON object, nothing else.
250 """
```

Classification Metrics

Precision: How many predicted positives were actually correct?

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

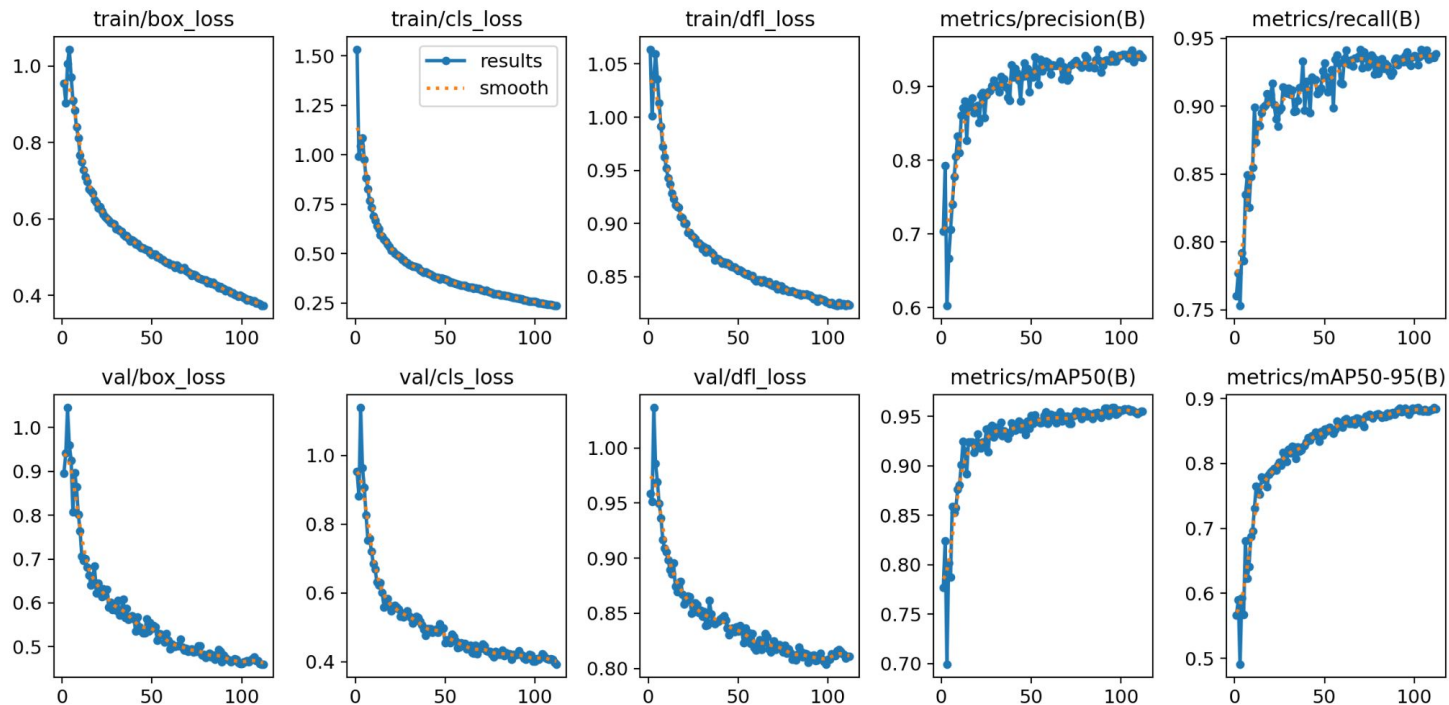
Recall: Of all the relevant items, how many did the model successfully find?

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

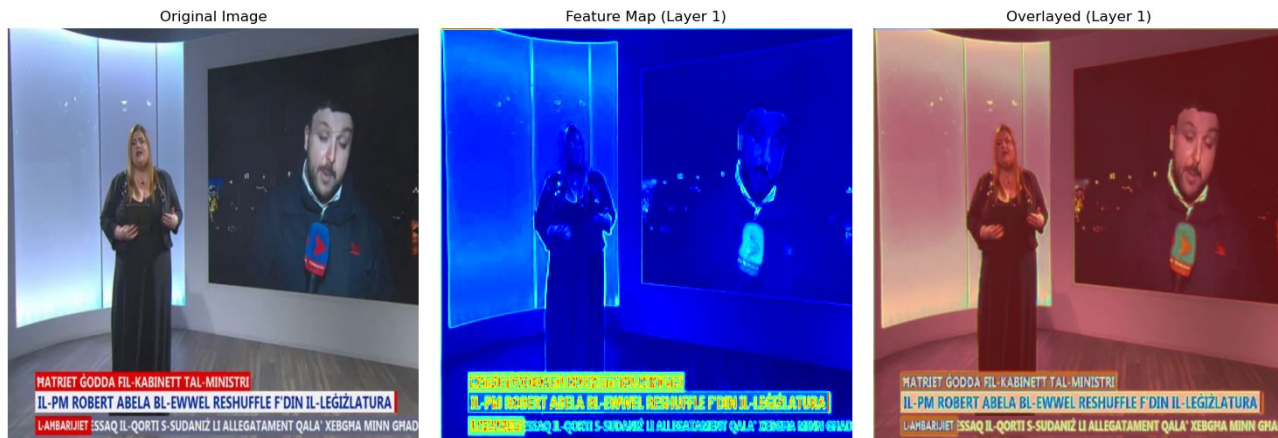
F1 Score: A harmonic mean of Precision and Recall – balances both concerns.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Model Performance and Loss Trends Over Epochs



Understanding Model Focus with Grad-CAM



Grad-CAM highlights which parts of the image the model focused on when making a prediction.

Strong focus on the ticker and name area, proving the model's attention is correct.